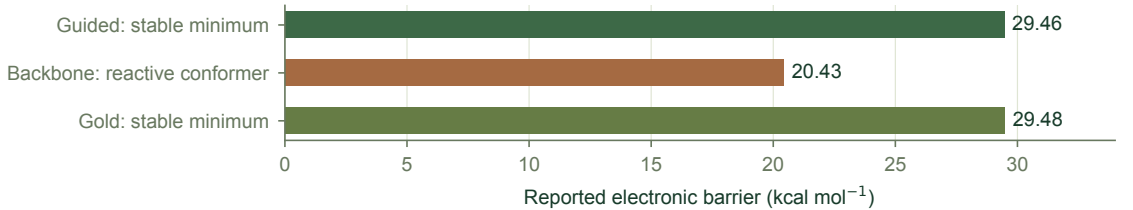
**a** **HA8: the reference state changes the barrier**




**b** **HH1: the same barrier, different evidence closure**




**c** **NU6: an energy difference needs a pathway identity**

**Concerted TS −247.030042392 Ha**

Betaine INT −247.060060699 Ha

**Difference = 18.836772 kcal mol$^{-1}$**

Guided: concerted saddle region; target INT → P saddle unresolved

Backbone: step-2 closure = 18.837